

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester IV)
EnterPrise Java
Practical-VI

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Class:TYIT	Batch:01
Date of Assignment:22-08-2026	Date/Time of Submission:22-08-2026

5. Implement the following Intermediate JSP Applications.

A.Develop a JSP application to demonstrate the use of JSP Directives and Implicit Objects.

Code(index.html):-

```
<%@ page language="java" contentType="text/html; charset=UTF-8" %>
<%@ page import="java.util.Date" %>
<html>
<head>
<title>JSP Directives and Implicit Objects</title>
</head>
<body>
<h1> Student Information </h1>
<h3> Nandini </h3>
<h3> Roll No.T006 </h3>
<h2>JSP Directives and Implicit Objects</h2>
<h3>JSP Directives</h3>
<p>Language: Java</p>
<p>Content Type: <%= response.getContentType() %></p>
<p>Current Date: <%= new Date() %></p>
<h3>Implicit Objects</h3>
<p>1. Request Method: <%= request.getMethod() %></p>
<p>2. Response Content Type: <%= response.getContentType() %></p>
<p>3. Session ID: <%= session.getId() %></p>
<p>4. Application Server: <%= application.getServerInfo() %></p>
<p>5. Page Name: <%= page.getClass().getName() %></p>
<p>6. PageContext: <%= pageContext.getClass().getName() %></p>
<p>7. Request Parameter: <%= request.getParameter("name") %></p>
<p>8. Config Servlet Name: <%= config.getServletName() %></p>
<p>9. Exception Object: Exception object is available only in error pages.</p>
</body>
</html>
```

Output:-

Student Information Nandini Roll No.T006 JSP Directives and Implicit Objects JSP Directives Language: Java Content Type: text/html; charset=UTF-8 Current Date: Sat Aug 22 11:26:48 IST 2026	Implicit Objects 1. Request Method: GET 2. Response Content Type: text/html; charset=UTF-8 3. Session ID: 80b0d36db58d15c7343a3d351816 4. Application Server: GlassFish Server Open Source Edition 4.1.1 5. Page Name: org.apache.jsp.index_jsp 6. PageContext: org.apache.jasper.runtime.PageContextImpl 7. Request Parameter: null 8. Config Servlet Name: jsp 9. Exception Object: Exception object is available only in error pages.
---	--

B.Develop a JSP application to demonstrate Form Processing and Input Validation.**Code(index.html):-**

```
<!DOCTYPE html>
<html>
<head>
  <title>Employee Registration Form</title>
</head>
<body>
  <h2>Employee Registration Form</h2>
  <form action="process.jsp" method="post">
    Employee Name:
    <input type="text" name="name">
    <br><br>
    Age:
    <input type="text" name="age">
    <br><br>
    Email:
    <input type="text" name="email">
    <br><br>
    Mobile:
    <input type="text" name="mobile">
    <br><br>
    Gender:
    <input type="radio" name="gender" value="Male"> Male
    <input type="radio" name="gender" value="Female"> Female
    <br><br>
    Department:
    <select name="department">
      <option value="">Select Department</option>
      <option value="IT">IT</option>
      <option value="HR">HR</option>
      <option value="Finance">Finance</option>
      <option value="Marketing">Marketing</option>
    </select>
    <br><br>
    Skills:
    <input type="checkbox" name="skill" value="Java"> Java
    <input type="checkbox" name="skill" value="Python"> Python
    <input type="checkbox" name="skill" value="HTML"> HTML
    <input type="checkbox" name="skill" value="SQL"> SQL
    <br><br>
    Experience:
    <select name="experience">
      <option value="">Select Experience</option>
      <option value="Fresher">Fresher</option>
      <option value="1-2 Years">1-2 Years</option>
      <option value="3-5 Years">3-5 Years</option>
      <option value="5+ Years">5+ Years</option>
    </select>
    <br><br>
    Address:
```

```

        <textarea name="address"></textarea>
        <br><br>
        <input type="submit" value="Register">
    </form>
</body>
</html>
Code(process.jsp):-
<%@ page contentType="text/html;charset=UTF-8" %>
<!DOCTYPE html>
<html>
<head>
    <title>Employee Details</title>
</head>
<body>
<h2>Employee Details</h2>
<%

```

```

    String name = request.getParameter("name");
    String age = request.getParameter("age");
    String email = request.getParameter("email");
    String mobile = request.getParameter("mobile");
    String gender = request.getParameter("gender");
    String department = request.getParameter("department");
    String experience = request.getParameter("experience");
    String address = request.getParameter("address");
    String[] skills = request.getParameterValues("skill");
    boolean valid = true;
    if (name == null || name.trim().isEmpty()) {
        out.println("Employee Name is required.<br>");
        valid = false;
    }
    if (age == null || age.trim().isEmpty()) {
        out.println("Age is required.<br>");
        valid = false;
    }
    if (email == null || email.trim().isEmpty()) {
        out.println("Email is required.<br>");
        valid = false;
    }
    if (mobile == null || mobile.trim().isEmpty()) {
        out.println("Mobile number is required.<br>");
        valid = false;
    }
    if (gender == null || gender.trim().isEmpty()) {
        out.println("Please select gender.<br>");
        valid = false;
    }
    if (department == null || department.trim().isEmpty()) {
        out.println("Please select a department.<br>");
        valid = false;
    }
    if (skills == null || skills.length == 0) {

```

```

        out.println("Please select at least one skill.<br>");
        valid = false;
    }
    if (experience == null || experience.trim().isEmpty()) {
        out.println("Please select experience.<br>");
        valid = false;
    }
    if (address == null || address.trim().isEmpty()) {
        out.println("Address is required.<br>");
        valid = false;
    }
    if (valid) {
%>
<h3>Employee Registration Successful</h3>
Employee Name: <%= name %>
<br><br>
Age: <%= age %>
<br><br>
Email: <%= email %>
<br><br>
Mobile: <%= mobile %>
<br><br>
Gender: <%= gender %>
<br><br>
Department: <%= department %>
<br><br>
Skills:
<%
    for (String skill : skills) {
        out.print(skill + " ");
    }
%>
<br><br>
Experience: <%= experience %>
<br><br>
Address: <%= address %>
<br><br>
<%
    }
%>
</body>
</html>

```

C.Develop a JSP application to display data from a MySQL Database using JDBC.

Code(index.jsp):-

```

<%@ page contentType="text/html; charset=UTF-8" pageEncoding="UTF-8" %>
<%@ page import="java.sql.Connection" %>
<%@ page import="java.sql.DriverManager" %>
<%@ page import="java.sql.Statement" %>
<%@ page import="java.sql.ResultSet" %>
<!DOCTYPE html>
<html>

```

Output:-

Employee Registration Form

Employee Name:

Age:

Email:

Mobile:

Gender: ☐ Male ☒ Female

Department:

Skills: ☐ Java ☒ Python ☒ HTML ☐ SQL

Experience:

Address:

Employee Details

Employee Registration Successful

Employee Name: Nandini Chaudhary

Age: 19

Email: nandini123@gmail.com

Mobile: 9876543267

Gender: Female

Department: IT

Skills: Python HTML

Experience: Fresher

Address: Andheri east

```

<head>
  <meta charset="UTF-8">
  <title>E-Commerce Product Details</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      background-color: #f2f2f2;
      margin: 0;
      padding: 30px;
    }
    h1 {
      text-align: center;
      color: #333;
    }
    .container {
      width: 95%;
      margin: auto;
      background-color: white;
      padding: 20px;
      border-radius: 10px;
      box-shadow: 0 0 10px #ccc;
    }
    table {
      width: 100%;
      border-collapse: collapse;
    }
    th {
      background-color: #333;
      color: white;
      padding: 12px;
      border: 1px solid #555;
    }
    td {
      padding: 10px;
      text-align: center;
      border: 1px solid #ddd;
    }
    tr:nth-child(even) {
      background-color: #f9f9f9;
    }
    tr:hover {
      background-color: #eee;
    }
    .price {
      color: green;
      font-weight: bold;
    }
    .discount {
      color: red;
      font-weight: bold;
    }

```

Output:-

```

mysql> USE ecommerce;
Database changed
mysql>
mysql> CREATE TABLE product (
  ->   product_id INT PRIMARY KEY AUTO_INCREMENT,
  ->   product_name VARCHAR(100) NOT NULL,
  ->   category VARCHAR(50) NOT NULL,
  ->   brand VARCHAR(50) NOT NULL,
  ->   price DECIMAL(10,2) NOT NULL,
  ->   quantity INT NOT NULL,
  ->   rating DECIMAL(2,1),
  ->   discount DECIMAL(5,2),
  ->   description VARCHAR(255)
  -> );

```

```

mysql> INSERT INTO product
  -> (product_name, category, brand, price, quantity, rating, discount, description)
  -> VALUES
  -> ('iPhone 15', 'Mobile', 'Apple', 69999.00, 10, 4.8, 10.00,
  -> 'Latest Apple smartphone with advanced camera'),
  -> ('Galaxy S24', 'Mobile', 'Samsung', 74999.00, 15, 4.7, 12.00,
  -> 'Premium smartphone with AMOLED display'),
  -> ('Inspiron 15', 'Laptop', 'Dell', 58999.00, 8, 4.5, 8.00,
  -> 'Powerful laptop for work and study'),
  -> ('WH-1000XM5', 'Headphones', 'Sony', 29999.00, 20, 4.9, 15.00,
  -> 'Wireless noise cancelling headphones'),
  -> ('Mechanical Keyboard', 'Keyboard', 'Logitech', 4999.00, 25, 4.6, 5.00,
  -> 'Mechanical keyboard with RGB lighting');

```

product_id	product_name	category	brand	price	quantity	rating	discount	
1	iPhone 15	Mobile	Apple	69999.00	10	4.8	10.00	
2	Galaxy S24	Mobile	Samsung	74999.00	15	4.7	12.00	
3	Inspiron 15	Laptop	Dell	58999.00	8	4.5	8.00	
4	WH-1000XM5	Headphones	Sony	29999.00	20	4.9	15.00	
5	Mechanical Keyboard	Keyboard	Logitech	4999.00	25	4.6	5.00	
6	iPhone 15	Mobile	Apple	69999.00	10	4.8	10.00	
7	Galaxy S24	Mobile	Samsung	74999.00	15	4.7	12.00	
8	Inspiron 15	Laptop	Dell	58999.00	8	4.5	8.00	
9	WH-1000XM5	Headphones	Sony	29999.00	20	4.9	15.00	
10	Mechanical Keyboard	Keyboard	Logitech	4999.00	25	4.6	5.00	

E-Commerce Product Details

Product ID	Product Name	Category	Brand	Price	Quantity	Rating	Discount	Description
1	iPhone 15	Mobile	Apple	Rs. 69999.0	10	4.8 / 5	10.0%	Latest Apple smartphone with advanced camera
2	Galaxy S24	Mobile	Samsung	Rs. 74999.0	15	4.7 / 5	12.0%	Premium Samsung smartphone with AMOLED display
3	Inspiron 15	Laptop	Dell	Rs. 58999.0	8	4.5 / 5	8.0%	Powerful laptop for work and study
4	WH-1000XM5	Headphones	Sony	Rs. 29999.0	20	4.9 / 5	15.0%	Wireless noise cancelling headphones
5	Mechanical Keyboard	Keyboard	Logitech	Rs. 4999.0	25	4.6 / 5	5.0%	Mechanical keyboard with RGB lighting
6	iPhone 15	Mobile	Apple	Rs. 69999.0	10	4.8 / 5	10.0%	Latest Apple smartphone with advanced camera
7	Galaxy S24	Mobile	Samsung	Rs. 74999.0	15	4.7 / 5	12.0%	Premium smartphone with AMOLED display
8	Inspiron 15	Laptop	Dell	Rs. 58999.0	8	4.5 / 5	8.0%	Powerful laptop for work and study
9	WH-1000XM5	Headphones	Sony	Rs. 29999.0	20	4.9 / 5	15.0%	Wireless noise cancelling headphones
10	Mechanical Keyboard	Keyboard	Logitech	Rs. 4999.0	25	4.6 / 5	5.0%	Mechanical keyboard with RGB lighting

```

        .rating {
            color: orange;
            font-weight: bold;
        }
        .description {
            text-align: left;
        }
        .error {
            color: red;
            font-weight: bold;
            text-align: center;
        }
    }
</style>
</head>
<body>
    <h1>E-Commerce Product Details</h1>
    <div class="container">
        <table>
            <tr>
                <th>Product ID</th>
                <th>Product Name</th>
                <th>Category</th>
                <th>Brand</th>
                <th>Price</th>
                <th>Quantity</th>
                <th>Rating</th>
                <th>Discount</th>
                <th>Description</th>
            </tr>
            <%
                Connection con = null;
                Statement stmt = null;
                ResultSet rs = null;
                try {
                    Class.forName("com.mysql.jdbc.Driver");
                    con = DriverManager.getConnection(
                        "jdbc:mysql://localhost:3306/ecommerce",
                        "root",
                        "root"
                    );
                    stmt = con.createStatement();
                    rs = stmt.executeQuery("SELECT * FROM product");
                    while (rs.next()) {
            %>
            <tr>
                <td><%= rs.getInt("product_id")%></td>
                <td><%= rs.getString("product_name")%></td>
                <td><%= rs.getString("category")%></td>
                <td><%= rs.getString("brand")%></td>
                <td class="price">
                    Rs. <%= rs.getDouble("price")%>

```

```

        </td>
        <td><%= rs.getInt("quantity")%></td>

        <td class="rating">
            <%= rs.getDouble("rating")%> / 5
        </td>
        <td class="discount">
            <%= rs.getDouble("discount")%>%
        </td>
        <td class="description">
            <%= rs.getString("description")%>
        </td>
    </tr>
    <%
    }

    } catch (Exception e) {
    %>
    <tr>
        <td colspan="9" class="error">
            Database Error: <%= e.getMessage()%>
        </td>
    </tr>
    <%
    } finally {
        try {
            if (rs != null) {
                rs.close();
            }
        } catch (Exception e) {
        }
        try {
            if (stmt != null) {
                stmt.close();
            }
        } catch (Exception e) {
        }
        try {
            if (con != null) {
                con.close();
            }
        } catch (Exception e) {
        }
    }
    %>
</table>
</div>
</body>
</html>

```